Module 3 Lab 1: PII Redaction Pipeline with Amazon Comprehend

20 points. Estimated time 90 to 120 minutes.

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Date: 11/4/26

How to use this file. Type your answers straight into the shaded boxes. Do not delete the questions. Anything marked as written in advance has to be filled in before you run that step, not afterwards. When you are finished, save this file and upload it with the rest of your submission.

1. Your unique values

These are what make the submission yours. They come from your own account and your own run, so no two students have the same ones.

Your Region: US-East-2

Your S3 bucket name: module3l1

The lowest confidence score you saw on any detection:
0.99+

Your chosen confidence cutoff and the one sentence reason for it:
My cutoff: 0.99+
Detections it drops: below .99
Why this number: I would want to be sure that no personal information can possibly be used maliciously, and if information that should not be redacted is redacted then we can fix that more easily compared to patients getting compromised.

The line number where something personal was still visible, if you found one:
11

2. Predictions, written in advance

Write each prediction BEFORE you run that step. A wrong prediction reasoned honestly earns full credit. One edited afterwards to match the result earns none.

Step 1

Before you choose Analyze, predict this: will the Entities tab list MORE items, FEWER items, or the SAME items as the PII tab for this ticket?

Same

What actually happened

They both had three items.

Why it was different, or why I was right

I was right with both showing three items, as when I looked at the text there were three main items within it which could be either an identifier or personal information.

3. Break it on purpose

Working code shows you what happens. Broken code shows you why. Do each of these deliberately, paste the real error text, and undo the change before moving on.

Break 1. Create a third Redactions job, but under PII entity types select only EMAIL. Run it and open the output.

Paste the exact error text here, not a description of it

No Error

What is now readable that should not be. Name the types that came through and say what a support agent could do with them.

Names, phone numbers, addresses, ticket numbers, and credit card numbers are now visible. This is dangerous if someone wants to steal information.

Break 2. In Real time analysis, paste a ticket but rewrite the person's name in all lowercase with no punctuation, for example sam reyes. Analyze it again.

Paste the exact error text here, not a description of it

Hi, this is sam reyes. My order has not arrived. You can reach me at sam.reyes@example.com or 310-555-5787.

This is the ticket analysed

Whether NAME was still found, and its confidence score before and after. Write both numbers down.

Name was still found with a 0.99+ confidence score both times.

Break 3. Create a job whose Output data location points at a bucket that is not yours, for example a name you invent.

Paste the exact error text here, not a description of it

OUTPUT_BUCKET_NOT_IN_SERVICE_REGION: The provided output S3 bucket is not in the service region.

The exact status the job list shows and the exact error text on the job detail page. Do not guess at it, read it.

Failed

4. Results

The numbers and outputs your own run produced.

Your Region and your S3 bucket name

US-East-2, bucket: module3l1

Your detection list from Steps 2 and 3, with confidence scores

- Step 2:

Entity	Type	Confidence
Sam Reyes	Name	0.99+
Sam.reyes@example.com	Email	0.99+
310-555-5787	Phone	0.99+

Switching to Labels view changes to just showing Types and Confidence
Types: Email, Name, Phone
Confidence: 1.00, 1.00, 1.00
I have no beginning and ending numbers.

Step 3:

Entity	Type	Confidence
dana.kim@example.com	Email	0.99+
Dana Kim	Name	0.99+
310-555-9558	Phone	0.99+
Lena Reyes	Name	0.99+
2523	Credit debit number	0.96
lena.reyes@example.com	Email	0.99+
Lena Nguyen	Name	0.99+
1906 Ocean Ave	Address	0.99+
310-555-9010	Phone	0.99+

Your chosen confidence cutoff and its one sentence reason

My cutoff: 0.99+
Detections it drops: below .99
Why this number: I would want to be sure that no personal information can possibly be used maliciously, and if information that should not be redacted is redacted then we can fix that more easily compared to patients getting compromised.

Your six count lines from Step 6

Then report:
lines checked 12
key says these contain personal data 10 contain personal data
fully redacted 9
something personal still visible yes line 11
clean line with something redacted yes line 12

5. Reflection

Your own words. No AI assistance on this part. Graders check these answers against the rest of your submission.

Question 1

Paste the redacted text of the one line where something personal was still visible, or if everything was redacted, the line with the lowest confidence score you saw. Explain what about that specific text made it hard for the service, and name one concrete change you would make to catch it.

Following up on ticket 4412. [NAME], [EMAIL]. Still no refund.

{"Entities": [{"BeginOffset": 15, "EndOffset": 21, "Type": "ADDRESS", "Score": 0.9999958276922101}], "File": "tickets.txt", "Line": 11}

The service did not see the ticket number as being an additional PII entity to redact.
Add a custom filter to add the ticket number as being a PII so it correctly gets redacted.

Question 2

Defend your confidence cutoff from Step 3 as though a healthcare compliance officer is reading it. Name the harm of a missed detection and the harm of a false one, say which you weighted more heavily, and quote your own lowest confidence score as part of the argument.

I still defend my confidence score of anything below 0.99+ being the cutoff. The harm of missing a detection is so much greater than a false one. If information is not caught that should be redacted, then identities can be stolen and lives ruined. Compared to if it was wrongfully redacted then a redo of the redaction sequence can happen to fix the small mistake.

6. AI use disclosure

Required by the course AI Use Policy.

Tool used (write none if you used no AI):

If you used one: what you asked it, what you kept, and how you checked it

7. Before you submit

- ☐ I checked twelve lines before I opened the answer key column
- ☐ My tickets.txt had sixty lines and no header
- ☐ My cutoff has a reason that names both harms
- ☐ I ran this entirely in the console and wrote no code
- ☐ Every prediction was written before I ran that step
- ☐ I recorded the actual result for every prediction, including the wrong ones
- ☐ I pasted the real error text for every break it item
- ☐ Work pushed to my GitHub portfolio repository and the URL is in here
- ☐ Reflection written by me with no AI assistance
- ☐ Files uploaded individually, no archive